

Principal Component Analysis (PCA) - Full Numerical Example (Step-by-Step)

Kinza 23K-0842

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Why Do We Use PCA?

PCA is widely used because real-world datasets often have:

- Redundant features (e.g., height and arm length)
- Correlated variables
- High dimensionality that slows ML models
- Noise that hides structure in data

PCA helps by:

- Removing correlations between features
- Reducing noise
- Compressing data
- Highlighting dominant patterns

Mathematical Intuition (Conceptual Overview)

1. Center the data by subtracting the mean.
2. Compute the covariance matrix.

3. Compute its eigenvalues and eigenvectors.
4. Sort eigenvalues from largest to smallest.
5. Eigenvectors corresponding to the largest eigenvalues become the principal components.
6. Project the data onto these components.

When Should PCA Be Used?

PCA is appropriate when:

- Features are correlated
- You want to visualize high-dimensional data
- You want to reduce noise
- You want to compress data before ML modeling
- You want to avoid the curse of dimensionality

PCA should **not** be used when:

- Features have different units (without scaling)
- Variables are categorical
- The dataset is highly nonlinear

Dataset

We consider the dataset with 5 observations and 2 features:

$$X = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 3 & 2 \\ 4 & 3 \\ 5 & 4 \end{bmatrix}$$

Let X_1 be the first feature and X_2 the second.

Step 1: Compute Feature Means

$$\bar{X}_1 = \frac{2 + 0 + 3 + 4 + 5}{5} = \frac{14}{5} = 2.8$$

$$\bar{X}_2 = \frac{0 + 1 + 2 + 3 + 4}{5} = \frac{10}{5} = 2$$

Step 2: Mean-Center the Data

We subtract the feature means from each observation:

$$X_{\text{centered}} = \begin{bmatrix} 2 - 2.8 & 0 - 2 \\ 0 - 2.8 & 1 - 2 \\ 3 - 2.8 & 2 - 2 \\ 4 - 2.8 & 3 - 2 \\ 5 - 2.8 & 4 - 2 \end{bmatrix} = \begin{bmatrix} -0.8 & -2 \\ -2.8 & -1 \\ 0.2 & 0 \\ 1.2 & 1 \\ 2.2 & 2 \end{bmatrix}$$

Step 3: Compute Covariance Matrix

The covariance matrix is:

$$C = \frac{1}{n-1} X_{\text{centered}}^T X_{\text{centered}}$$

Compute each entry:

$$\begin{aligned} C_{11} &= \frac{(-0.8)^2 + (-2.8)^2 + (0.2)^2 + (1.2)^2 + (2.2)^2}{4} \\ &= \frac{0.64 + 7.84 + 0.04 + 1.44 + 4.84}{4} = \frac{14.8}{4} = 3.7 \end{aligned}$$

$$C_{22} = \frac{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}{4} = \frac{4 + 1 + 0 + 1 + 4}{4} = \frac{10}{4} = 2.5$$

$$\begin{aligned} C_{12} = C_{21} &= \frac{(-0.8)(-2) + (-2.8)(-1) + (0.2)(0) + (1.2)(1) + (2.2)(2)}{4} \\ &= \frac{1.6 + 2.8 + 0 + 1.2 + 4.4}{4} = \frac{10}{4} = 2.5 \end{aligned}$$

Thus,

$$C = \begin{bmatrix} 3.7 & 2.5 \\ 2.5 & 2.5 \end{bmatrix}$$

Step 4: Eigenvalues of the Covariance Matrix

Solve:

$$\det(C - \lambda I) = 0$$

$$\begin{vmatrix} 3.7 - \lambda & 2.5 \\ 2.5 & 2.5 - \lambda \end{vmatrix} = 0$$

$$(3.7 - \lambda)(2.5 - \lambda) - 2.5^2 = 0$$

Expand:

$$(3.7)(2.5) - 3.7\lambda - 2.5\lambda + \lambda^2 - 6.25 = 0$$

$$9.25 - 6.2\lambda + \lambda^2 - 6.25 = 0$$

$$\lambda^2 - 6.2\lambda + 3 = 0$$

Solve quadratic:

$$\lambda = \frac{6.2 \pm \sqrt{6.2^2 - 4 \cdot 3}}{2}$$

$$= \frac{6.2 \pm \sqrt{38.44 - 12}}{2}$$

$$= \frac{6.2 \pm \sqrt{26.44}}{2}$$

$$\lambda_1 \approx \frac{6.2 + 5.142}{2} = 5.671$$

$$\lambda_2 \approx \frac{6.2 - 5.142}{2} = 0.529$$

Step 5: Eigenvectors

Solve:

$$(C - \lambda I)v = 0$$

For $\lambda_1 = 5.671$:

$$\begin{bmatrix} 3.7 - 5.671 & 2.5 \\ 2.5 & 2.5 - 5.671 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1.971 & 2.5 \\ 2.5 & -3.171 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

Row 1:

$$-1.971v_1 + 2.5v_2 = 0$$

$$v_2 = 0.7884v_1$$

Normalize:

$$\begin{aligned} v &= \begin{bmatrix} 1 \\ 0.7884 \end{bmatrix} \Rightarrow \frac{1}{\sqrt{1 + 0.7884^2}} \begin{bmatrix} 1 \\ 0.7884 \end{bmatrix} \\ &= \frac{1}{1.270} \begin{bmatrix} 1 \\ 0.7884 \end{bmatrix} = \begin{bmatrix} 0.787 \\ 0.620 \end{bmatrix} \end{aligned}$$

Thus PC1 eigenvector:

$$e_1 = \begin{bmatrix} 0.787 \\ 0.620 \end{bmatrix}$$

Similarly, PC2 is:

$$e_2 = \begin{bmatrix} -0.620 \\ 0.787 \end{bmatrix}$$

Step 6: Variance Explained

$$\text{Total variance} = 5.671 + 0.529 = 6.2$$

$$\text{Explained by PC1} = \frac{5.671}{6.2} = 91.47\%$$

$$\text{Explained by PC2} = \frac{0.529}{6.2} = 8.53\%$$

Step 7: Project Data onto PC1

Projection:

$$Z = X_{\text{centered}}e_1$$

Compute each point:

1st point:

$$z_1 = (-0.8)(0.787) + (-2)(0.620) = -0.6296 - 1.24 = -1.8696$$

2nd:

$$z_2 = (-2.8)(0.787) + (-1)(0.620) = -2.2036 - 0.62 = -2.8236$$

3rd:

$$z_3 = (0.2)(0.787) + (0)(0.620) = 0.1574$$

4th:

$$z_4 = (1.2)(0.787) + (1)(0.620) = 0.9444 + 0.62 = 1.5644$$

5th:

$$z_5 = (2.2)(0.787) + (2)(0.620) = 1.7314 + 1.24 = 2.9714$$

Final projected PC1 coordinates:

$$Z = \begin{bmatrix} -1.8696 \\ -2.8236 \\ 0.1574 \\ 1.5644 \\ 2.9714 \end{bmatrix}$$

Final Interpretation

- PC1 explains $\approx 91.5\%$ of total variance.
- PC1 is a weighted combination of both features:

$$\text{PC1} = 0.787X_1 + 0.620X_2$$

- Data projected on PC1 shows clear ordering from low to high values.
- PC2 contains only 8.5% of the variance, mostly noise for this dataset.

Advantages of PCA

- Reduces dimensionality while preserving information
- Removes multicollinearity
- Reduces computational cost for ML algorithms
- Helps visualize high-dimensional data
- Improves model accuracy by removing noise

Disadvantages of PCA

- Principal components are hard to interpret
- PCA is sensitive to feature scaling
- PCA assumes linear relationships only
- PCA may discard useful information if small-variance features matter
- Components are combinations of original features, not physical variables

Applications of PCA

- Face recognition and image compression
- Genome data and bioinformatics
- Finance (risk management, portfolio reduction)
- Pattern recognition and signal processing
- Exploratory Data Analysis (EDA)
- Noise reduction in sensors and IoT